

CHAPTER 1

INTRODUCTION

The main objective is to design speed control and automatic braking system in electric vehicles to automatically reduce speed or to apply brakes if the driver fails to react in time when the vehicle in front slows down or stops. The primary function of automatic braking system is to mitigate the impact or avoid collisions during critical situations by applying the brakes autonomously. Usual road accidents happen when drivers applies inadequate, delayed or absolutely no brakes at all to avoid collisions. The most common reasons why drivers fail to step on the brakes at the right time are distractions,inattentiveness,sleepiness,visibility conditions ,weather conditions ,etc. Automatic braking system is specifically designed to function properly on different sets of road scenarios. First, it alerts the driver of existing obstacles in front of a vehicle. If the driver fails to act accordingly on time to avoid an impact, the system will automatically reduce the speed of the vehicle with different levels of braking force to avoid or at least lessen the impact of the collision.

Under certain circumstances like a sudden obstacle entering from the side of the road, the driver will not be able to react and apply brakes at all, in such cases collision with the obstacle cannot be avoided.In a vehicle with automatic braking system the vehicle will automatically attempt to reduce the velocity and thereby reducing the momentum and reduces the overall damage due to the collision.[6]

CHAPTER 2

LITERATURE REVIEW

Automatic braking is a technology for automobiles to sense an imminent collision with another vehicle, person or obstacle and to apply brakes to slow the vehicle without any driver input. Sensors are used to detect other vehicles or obstacles they can be radar, video, infrared, laser, ultrasonic or other technologies.

There is a threshold safe distance calculated by the system and if the driver fails to respond even when the vehicle crosses that region, then only brakes will be applied automatically. Many vehicles are provided with the option of turning on or off the automatic system based on their surroundings. In some automobiles even though they cannot be completely disabled, they can be limited to warning the driver about coming obstacle. Even this emergency braking initiates ABS which help the driver to retain the control over vehicle without any skidding. Automatic braking system is only effective if the mode of sensing the obstacles is reliable, or else any kind of false interpretation may cause a lot of damage.

An emergency braking system (AEBS) or autonomous emergency braking (AEB) is an autonomous road vehicle safety system which employs sensors to monitor the proximity of vehicles in front and detects situations where the relative speed and distance between the host and target vehicles suggest that a collision is imminent. In such a situation, emergency braking can be automatically applied to avoid the collision or at least to mitigate its effects. United Nations Economic Commission for Europe(UNECE) has announced that this kind of system will become mandatory for new heavy vehicles. A recent study suggests that if all cars feature the system, it will reduce accidents by up to 27 percent and save up to 8000 lives per year.

Vehicle technology has increased rapidly in recent years, particularly in relation to braking systems and sensing systems. The widespread introduction of anti-lock braking systems (ABS) has provided the building blocks for a wide variety of braking control systems. Additional hardware that allows brake pressure to be increased above pedal demand as well as to be reduced, combined with additional software control algorithms and sensors allow traction control (TC), electronic brake force distribution (EBD), brake assist (BA) and electronic stability control (ESC) functions to be added.

In parallel to the development of braking technologies, sensors have been developed that are capable of detecting physical obstacles, other vehicles or pedestrians around the vehicle. Many luxury, mid- size and small cars in Europe, and in Japan even very small cars (Daihatsu Move), are now fitted with an adaptive cruise control (ACC) system that is capable of measuring and maintaining a driver-preset headway to the vehicle ahead by automatic modulation of the engine control, and if required, automatically applying brakes up to a maximum deceleration of 0.3g (as per ISO standard). If no vehicle is ahead, the vehicle maintains the desired “set-speed”. ACC can be ordered as an option for new vehicles. At least three heavy truck manufacturers offer this feature on their vehicles.

Theoretically, a vehicle equipped with modern braking technology and adaptive cruise control is equipped with the basic building blocks for a simple (braking only – no steering) collision avoidance system that would be capable of detecting when a collision is likely to occur and applying emergency braking to avoid it. More advanced and/or multiple sensors are likely to be required as well as considerable further development before full collision avoidance systems (other than low speed systems) are available in production. However, collision mitigation systems are already on the market, providing limited automated braking capability and some systems are available that can automatically avoid collisions in low speed traffic.

Automatic emergency braking systems (AEBS) were in production on a number of current vehicles at the top end of the market and are capable of autonomously mitigating two-vehicle front to rear shunt accidents as well as some collisions with fixed objects and motorcycles. Such systems were fitted alongside ACC and forward collision warning systems that shared the same hardware.

Systems are currently in various phases of development that will also act in pedestrian collisions and towards the end of the project at least one system offering some pedestrian functionality was released on a production vehicle. There is also a strong research base that aims to develop systems capable of acting in other vehicle to vehicle impact configurations.

Clear functional requirements for AEBS are in existence in Japan and appear to be appropriate to use as guidance for systems in Europe subject to modification of some limit values. An ISO standard is under development but was not yet available for study. There remains insufficient information available at this time to produce more rigorous standards that more closely define performance or to define methods of testing the effectiveness of the whole system and further research may be required in this area.

Further requirements are required to cover the integrity of the system. Some of these, such as reliability of the system and resistance to EMC issues are already embodied in regulation and are covered, at least to a minimum standard. There was however, evidence to suggest that further research was required to examine whether the safety critical nature of AEBS requires more stringent limits in these areas. Other areas such as environmental protection including immunity to electro-static discharge and particularly relating to the compatibility and reliability of similar and different systems when numerous modules are on the road are not covered and may need additional measures. In particular, the possibility that interference could occur when multiple sensing systems “meet” at a busy road section may require further investigation.

Substantial difficulties have been encountered in trying to define the benefits of an AEBS in terms of casualty reduction. These are related to fundamental limitations in terms of the detail available in accident databases and the reconstruction methods used to generate them and also related to the limited scope of the project which considered only the benefits of the automated braking and not the benefits of ACC, the collision warning and optimised restraint functions.

It was not possible to establish detailed and accurate estimates of the costs of systems because of commercial sensitivity and difficulties in separating the costs of different functions that shared the same hardware. Only current retail prices of optional systems, target costs for future systems and a broad range of generic cost estimates from industry were available.

Instead of a detailed analysis of the costs and benefits of current, first generation AEBS, a high level scoping analysis was carried out to indicate the potential benefits if certain levels of effectiveness could be achieved in a wide range of collision situations. This analysis reached the following conclusions.

Immediate introduction of AEBS that approximated first generation system functions (front to rear collisions and collisions with rigid fixed objects located on the carriageway) to all light vehicles offered considerable casualty reduction potential but was unlikely to be cost effective unless substantial cost reductions could be achieved. The main reasons for this were the high frequency/low severity nature of light vehicle shunt accidents and the high number of vehicles that must be fitted with the technology to achieve the benefits.

Immediately equipping all new heavy vehicles with such a system would yield a lower casualty reduction but a better benefit to cost ratio because of the much lower number of vehicles requiring fitment and the increased severity of front to rear collisions involving heavy vehicles.

Introducing AEBS that was also effective in collisions with pedestrians and with fixed objects off the carriageway substantially changed the view of the systems. The casualty reduction potential in this situation was greatly increased and if fitted to all M and N vehicles it would be likely to have a fatality saving potential in the thousands if implemented across the EU vehicle fleet. The research suggests that at some point in the next three to eight years the technology to enable these functions will be readily available. The limited evidence available regarding the costs of the systems suggested alternative possibilities where either a combination of “low cost” sensors and increased production volumes decreased the cost or where increased technical complexity increased the costs. Achieving the considerable potential benefits with a substantially positive benefit to cost ratio is therefore strongly dependant on the system cost.

Further developments to allow function in head on and junction collisions also offer a substantial increase in both the casualty reduction potential and the benefit to cost ratio, although increased technical difficulty may increase costs or limit benefits.

Substantial technical difficulties are likely to be encountered when considering fitment of AEBS to motorcycles but if these can be overcome then substantial casualty savings are achievable with a high potential for a benefit to cost ratio in excess of one.

Overall, it was found that AEBS is highly likely to be a very effective safety measure in terms of both casualty reduction and benefit to cost ratio in the relatively near future, provided further technical development and cost reduction take place.

CHAPTER 3

PROPOSED SYSTEM

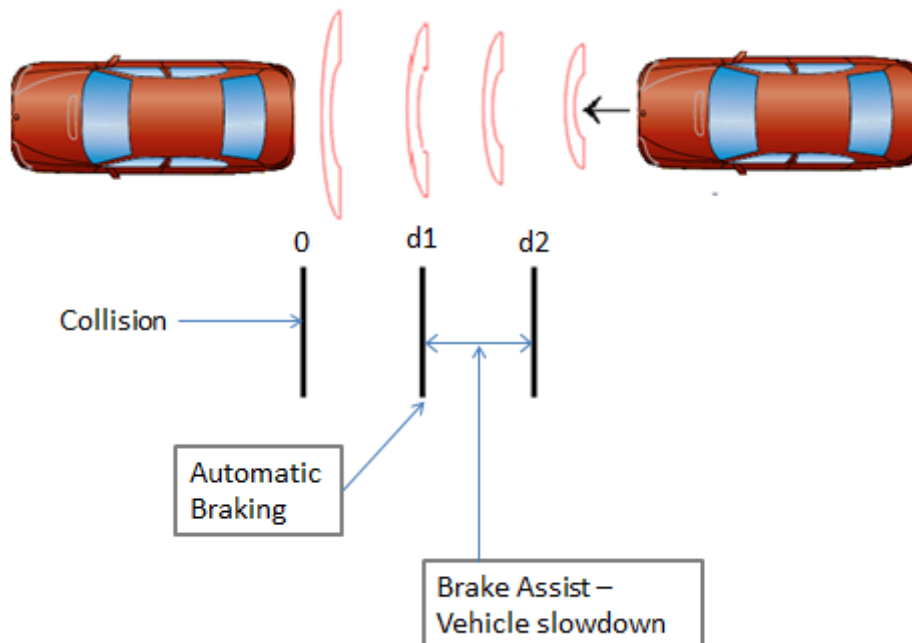


Fig 3.1: Basic Working

The proposed work is likely to control the speed of vehicle and automatically apply brakes. It is divided into three main steps;

1. Detect the obstacle.
2. Control the speed of vehicle.
3. Automatic braking system.

A sensor mounted on the vehicle is used to detect obstacles in front of it. If there is no obstacle the vehicle is free to accelerate and decelerate according to the driver's choice. When an obstacle is detected and if the distance between the obstacle and the vehicle keeps on decreasing and if the driver does not apply brakes due to lack of attention, the system takes over and decreases

the speed of the vehicle by using PWM technique to reduce the speed of the motor to avoid collision. This is done when the distance between the obstacle and the vehicle is a set value (say d_2). The slowing down of the vehicle continues until the distance becomes another set value (say d_1). During this period the driver is allowed to decelerate but acceleration of the vehicle is not possible. If the vehicle crosses the minimum distance (d_1) the system brings the vehicle to a complete stop by applying full brakes.

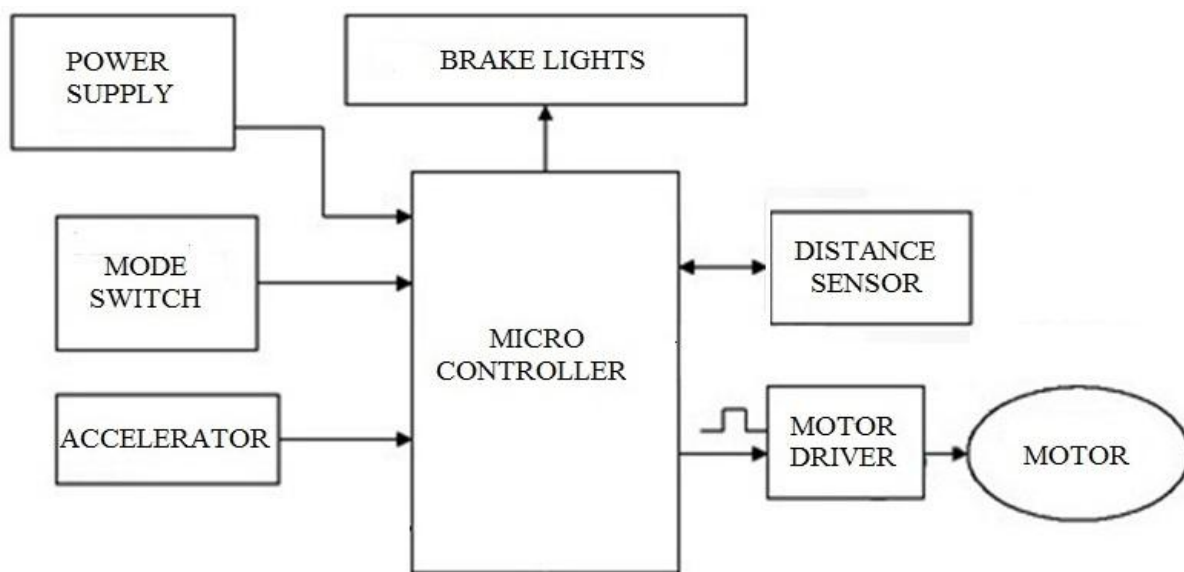


Fig 3.2: Block Diagram

The microcontroller used here is Arduino Uno. A potentiometer is provided to accelerate and decelerate the connected vehicle respectively. It has been equipped with Warning Lights to notify the driver the reducing distance between the vehicle and an obstacle. The motor of the vehicle is interfaced to the Arduino Uno Microcontroller using an L298N Motor Driver. An Ultrasonic Sensor has been connected to the Microcontroller to detect the presence of obstacles; it comprises of the ADC and the Distance Sensor. It has a Mode Switch which can be used to enable and disable the system when needed.

CHAPTER 4

STOPPING DISTANCE

Braking distance is the distance a vehicle will travel from the point when its brakes are fully applied to when it comes to a complete stop. It mainly depends on separation between the two vehicles, difference in their velocities and friction coefficient (ie. friction coefficient between the vehicle tires and the road) [6]. It is primarily affected by the original speed of the vehicle and the coefficient of friction between the tires and the road surface, and negligibly by the tires' rolling resistance and vehicle's air drag. The type of brake system in use only affects trucks and large mass vehicles, which cannot supply enough force to match the static frictional force.

Typical Stopping Distances

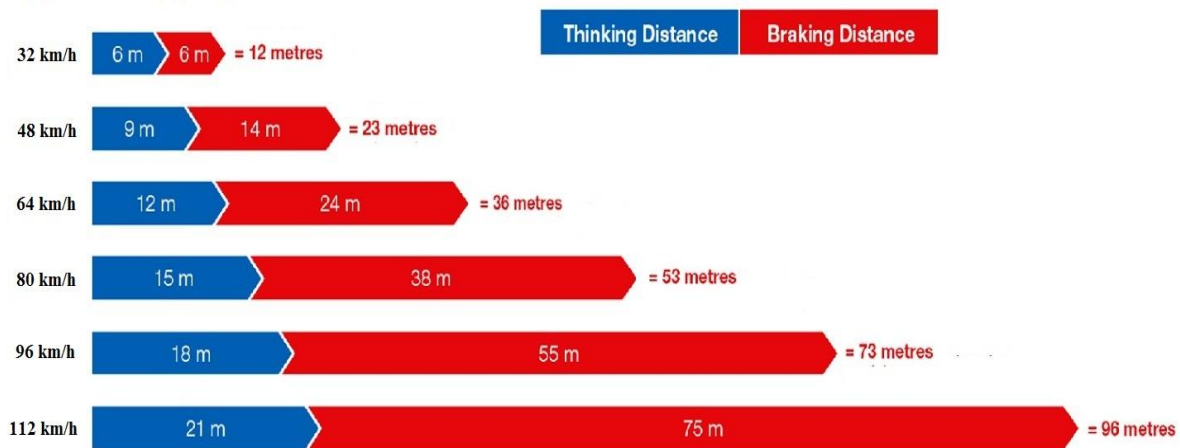


Fig 4.1: Typical stopping distances.

The braking distance is one of two principal components of the total stopping distance. The other component is the reaction distance or thinking distance, which is the product of the speed and the perception-reaction time of the driver. A perception-reaction time of 1.5 seconds, and a coefficient of kinetic friction of 0.7 are standard for the purpose of determining a bare baseline for accident reconstruction and judicial notice; most people can stop slightly sooner under ideal conditions.

Total stopping distance

The total stopping distance is the sum of the perception-reaction distance and the braking distance.

$$D_{total} = D_{p-r} + D_{braking} = vt_{p-r} + \frac{v^2}{2\mu g}$$

A common baseline value of

$$t_{p-r} = 1.5s, \mu = 0.7$$

is used in stopping distance charts. These values incorporate the ability of the vast majority of drivers under normal road conditions. However, a keen and alert driver may have perception-reaction times well below 1 second, and a modern car with computerized anti-skid brakes may have a friction coefficient of 0.9--or even far exceed 1.0 with sticky tires.

Experts historically used a reaction time of 0.75 seconds, but now incorporate perception resulting in an average perception-reaction time of: 1 second for population as an average; or even a 2.5 second reaction time—to specifically accommodate very elderly, debilitated, intoxicated, or distracted drivers. The coefficient of friction may be 0.25 or lower on wet or frozen asphalt, and anti-skid brakes and season specific performance tires may somewhat compensate for driver error and conditions.

Actual total stopping distance

The actual total stopping distance may differ from the baseline value when the road or tire conditions are substantially different from the baseline conditions or when the driver's cognitive function is superior or deficient. To determine actual total stopping distance, one would typically empirically obtain the coefficient of friction between the tire material and the exact road spot under the same road conditions and temperature. They would also measure the person's perception and reaction times.

By the implementation of the automatic braking system it is possible to reduce the total braking distance considerably as the reaction time of the system is almost immediate and the thinking distance becomes negligible.

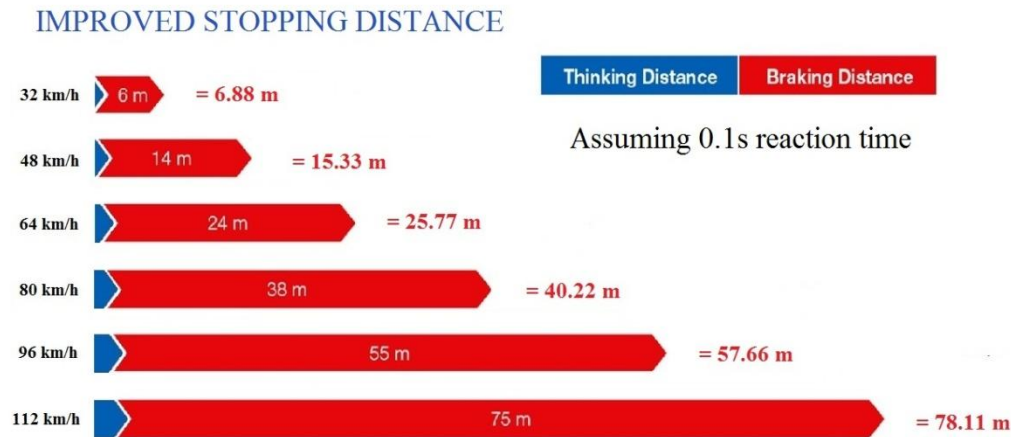


Fig 4.2:Improved stopping distances.

The improved stopping distance by implementing the system is shown in Figure .The thinking distance is considerably reduced,in turn the overall stopping distance is also reduced.

4.1 DISTANCE MEASUREMENT

The most important part of the system is to detect an obstacle such as a vehicle, person, animal etc. in order to sense an imminent collision.

Each car manufacturer has its own automatic braking system technology, but they all rely on some type of sensor input. Some of these systems use lasers, ultasonic technology, others use radar, and some even use stereoscopic cameras. This sensor input is then used to determine if there are any objects present in the path of the vehicle. If an object is detected, the system can then determine if the speed of the vehicle is greater than the speed of the object in front of it. A significant speed differential may indicate that a collision is likely to occur, in which case the system is capable of automatically activating the brakes[8].After detecting the obstacle neccesary commands and signals are passed to the control unit for futher action.

4.1.1 Radar sensors



Fig 4.3: Radar Sensor by Bosch.

Radar is currently the most widely adopted sensing technology for automotive ranging applications but careful design and optimisation are needed for a particular sensing task (e.g. frequency, antenna type and size, range, power limitations etc.) and frequency band availability is limited. It is suitable for both short and long range applications (typically <1 to 200m) and is superior to many other technologies for vector analysis. Radar sensors are also known to have sufficient robustness to function reliably under harsh environmental conditions for extended periods of time. Their performance is affected by atmospheric attenuation (particularly molecular oxygen and water droplets and precipitation will increase attenuation, especially at higher frequencies). Radar sensors are of medium to high cost and offer poor recognition of lane markings. They rely on objects reflecting the radar waves so objects with poor reflectivity are not easily detected.

The simplest type of radar system is a basic rangefinder that measures the time of flight of the transmitted radar beam from its reflection. This provides the distance between the sensor and the obstacle. Another type of radar uses the Doppler effect where the reflected beam undergoes a frequency shift proportional to the relative speed of the transmitter/receiver and the obstacle. Pulsed radar is frequently used in automotive applications to continuously update the position

and closing speed etc. of the obstacles in the radar field of view. Other techniques include frequency modulation and frequency shift keying. Combinations of several radar beams can be used to accurately triangulate the positions of objects within the vehicle foreground up to several hundred metres.

4.1.2 Passive infrared sensors



Fig 4.3: Infrared sensor by Allen Bradley.

Infrared radiation, more commonly identified as thermal radiation, is emitted by all objects having a temperature above absolute zero. A passive infrared sensor detects this naturally emitted radiation. Infrared radiation sits between the visible and microwave region of the electromagnetic spectrum and has a wavelength range of approximately 750nm to 1mm. Infrared sensors have numerous applications in object detection. Sensor functionality varies dramatically; high resolution (10 to 100,000 element) arrays with high sensitivity have been developed for military use, whilst volume manufactured one or two element detectors are found in intruder systems and simple light sensor applications. The pyroelectric detectors

respond to changes in thermal characteristics – objects that are moving relative to the sensor are detected by the thermal contrast between the object and its background.

In practical terms a passive infrared sensor can be as simple as a single semiconductor detector peaked to respond in the infrared band. Arrays of these types of sensor can be put together to form a crude infrared “camera”. Traditional cameras are also available designed to work at infrared wavelengths. Such designs are used by the military and police for surveillance applications. These systems allow the detection of humans at night and provide a complete night vision scenario.

Passive infrared sensors are compact, provide good object detection and are suitable for pedestrian detection, however low resolution sensors may prove inadequate at reliably identifying obstacles, whilst the increased cost of high resolution sensors may prohibit their use. They are typically poor at differentiating between multiple hot bodies in close proximity and detection can be influenced by the presence of thermal insulation. For example, a pedestrian wearing a thick coat may appear to be relatively cold. Performance is also affected by the emissivity of the detected object and environmental conditions; the emitted radiation being absorbed by particles in the air. The effective distance at which sensors can reliably identify obstacles in automotive applications has yet to be established.

4.1.3 Active infrared sensors

The receiver in an active infrared sensor performs in a similar manner to that of a passive device, albeit the sensed area is either flooded with infrared radiation or a scanning laser beam may be used. Advantages of active infrared sensors over passive devices include improved range and sharp directivity. They provide a high data transmission rate and good immunity to electromagnetic disturbances. They are typically poor at differentiating between multiple hot bodies. Performance is also affected by the reflectivity of the detected object and environmental conditions.

4.1.4 Laser sensors



Fig 4.3: Lidar sensor by Sick.

Laser or lidar (light detection and ranging) sensors are effectively range finders that calculate the distance to an object by measuring the time of flight of the emitted and reflected signal. In this respect, their principle of operation is similar to radar. Varying the direction of the emitted beam facilitates scanning over a wide area at high frequency to build up a map of points describing the environment and any potential obstacles. Frequently updating the map enables the determination of the relative position and speed of obstacles.

Laser scanner systems are capable of up to 100% detection accuracy in fair weather and 95% classification accuracy (Nuttall & Myers, 1996), overcoming many of the problems associated with radar sensors. However their performance is affected by output power, reflection, refraction, absorption and scattering. They provide good lateral resolution but are poor at recognising lane markings and require complex shape and motion recognition algorithms to be able to differentiate between different types of obstacles.

It is important that laser based systems are operated correctly because exposure to high power laser light can seriously damage human eyesight. This can be achieved by limiting the power and restricting the wavelength. Since laser systems require a clear line of sight they are usually mounted behind the windscreen whereas radar sensors can effectively ‘see through’ non-metallic structures.

4.1.5 Ultrasonic sensors

Like radar, lidar and active infrared systems, ultrasound can be used in detection and ranging applications using the time of flight principle to estimate the distance to an object. Ultrasonic emissions are effectively sound waves with frequencies higher than that audible to the human ear, suitable for short to medium range applications at low speed. A scanning sonar sensor based on a phased array of ultrasonic sensors facilitates gathering information on the distance, angular position, velocity and nature of surrounding obstacles. Ultrasonic sensors provide a good indication of vehicle to obstacle distances, are less susceptible to being affected by a build up of debris, have good response times and are low cost. However their performance is only suitable for short and medium range applications, fluctuations in operating voltage reduces performance and the accuracy of object detection is sometimes affected by reflected signals.

4.1.6 Imaging systems



Fig 4: Stereoscopic camera.

Cheap video cameras are now readily available and provide a low cost solution to image detection. However, the fundamental problem when using cameras is the need to distinguish particular objects within the image requiring complex software. Much research is going on in this field, for example the EC PReVENT project.

In their basic single camera configuration video imagers cannot provide ranging information. However, using the principle of triangulation coupled to obstacle recognition routines, the images viewed by camera systems may be processed to facilitate identification of obstacles and their range. Such systems provide good obstacle and environmental recognition using non-

intrusive means. Their performance can be affected by contamination of the external optical surface of the viewer and by air-borne emissions affecting vision (e.g. precipitation, fog, smoke etc.). The ability to process environmental data in real time is based upon a large number of image simplifying assumptions, which may prove to be insufficiently robust for identifying complex urban road environment and obstacle movements reliably. Imaging technology provides the most complete environmental information. A particular type of camera is the 3D imager. These devices include a laser emitter built into the camera that illuminates the scene. In effect, each pixel of the camera sensor acts as a simple 'Lidar like' range finder. Further details of this type of system are provided in the EC PReVENT project, sub-project UseRCams.

4.1.7 Passive mmWave sensors

Millimetre waves (mm-waves) are electromagnetic waves similar to light and radio waves. The name "mm-waves" is given to the region of the electromagnetic spectrum between frequencies of 30GHz and 300GHz, which correspond to wavelengths of 10 to 1mm. Mm-waves can be regarded either as very high frequency radio waves or as very long wavelength infrared. They should not be confused with the shorter wavelength terahertz (THz) waves though the terms are often interchanged.

All objects which are above absolute zero in temperature (-273 degrees C approximately) emit thermal radiation according to Planck's law and according to the emissivity of their surfaces. Human skin is highly emissive and its emission peaks in the infrared range at a wavelength of about 10 μ m. Infrared emissions are normally blocked by clothing which is worn with the express purpose of preventing heat loss. However, this clothing is largely transparent to millimetre-wave radiation. This offers the possibility of detecting human presence by their mm-wave emissions, a technique which is already in use by the military and security forces to detect the presence of enemy or undesirable personnel in buildings or vehicles.

There are two types of passive mmWave detectors that may be suitable for vehicle applications. These are imagers that behave like a camera except that they are tuned to a different part of the electromagnetic spectrum and discrete detectors that behave like a simple photo-cell, known as bolometers. These types of detector are being examined within the EC PreVENT project, sub-projects ProFusion and COMPOSE.

4.1.8 Sensor fusion

Sensor fusion involves combining multiple sensing technologies to supplement, enhance and improve the reliability of sensing system. Using one sensor technology to compensate for the drawbacks of another is a common practice in sensor fusion applications. A common fusion application involves the combination of radar or laser scanner and image sensing system. Sensor fusion offers the potential to gather additional information describing the vehicle operational environment and any obstacles but complex algorithms are necessary to interpret the data gathered.

4.2 Reliability of the sensing system

The reliability of a system is the ability to provide optimum performance within a range of operating conditions and over specified time duration. The reliability of an automotive sensing system will depend on a number of factors, including the reliability of components and subcomponents, the nature of interface between components and subcomponents, operational conditions and environmental interference, integrity of the software for data processing etc. There are many approaches to estimating systems reliability and different methods often yield varying estimates of reliability.

CHAPTER 5

ECU AND PWM BASED SPEED CONTROL

In electric vehicles the electric motors propelling the vehicle is controlled by an Electronic Control Unit (ECU) or a Power Control Unit (PCU). Modern E.C.U.s use a microprocessor which can process the inputs from the various sensors in real-time.

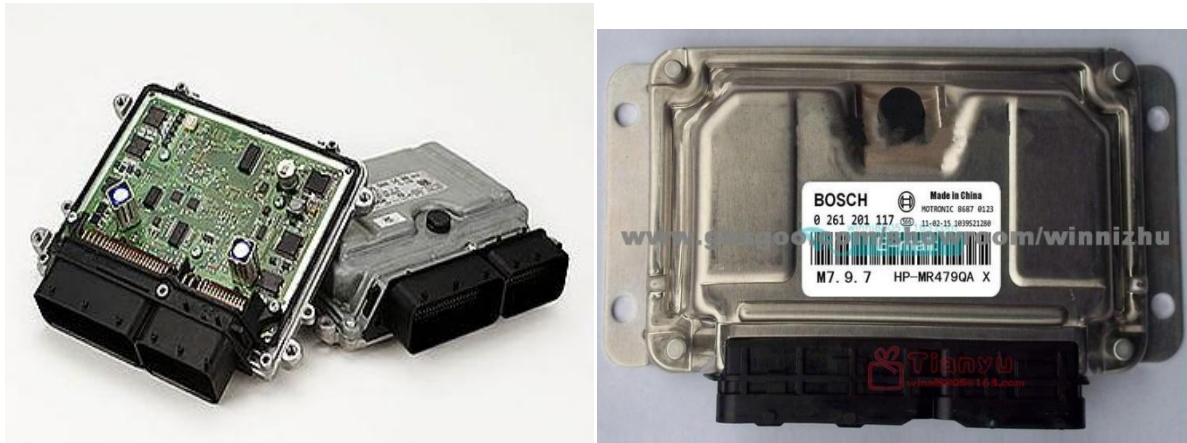


Fig 4.3: Automobile ECU.

An electronic control unit contains the hardware and software (firmware). The hardware consists of electronic components on a printed circuit board (P.C.B.), ceramic substrate or a thin laminate substrate. The main component on this circuit board is a micro controller chip (CPU). The software is stored in the microcontroller or other chips on the P.C.B., typically in E.P.R.O.Ms or flash memory so the C.P.U. can be re-programmed by uploading updated code or replacing chips. An electronic control unit controls power delivered to electric motor to ensure optimal vehicle performance. It does this by reading values from a multitude of sensors within the vehicle, interpreting the data using multidimensional performance maps (called lookup tables), and adjusting the parameters accordingly. The speed of the electric motor is controlled via Pulse Width Modulation (PWM). The ECU will increase or decrease the ON time and OFF time of the entire pulse time to vary the speed of the electric motor based on the accelerator input given by the driver. If we decrease the ON time then the voltage applied to the DC motor will reduce and the speed of the DC motor will be reduced. (ie. The duty ratio

of the PWM signal is varied). Duty ratio is the ratio of total ON time of the pulse to the sum of ON time and OFF time of the pulse[2].

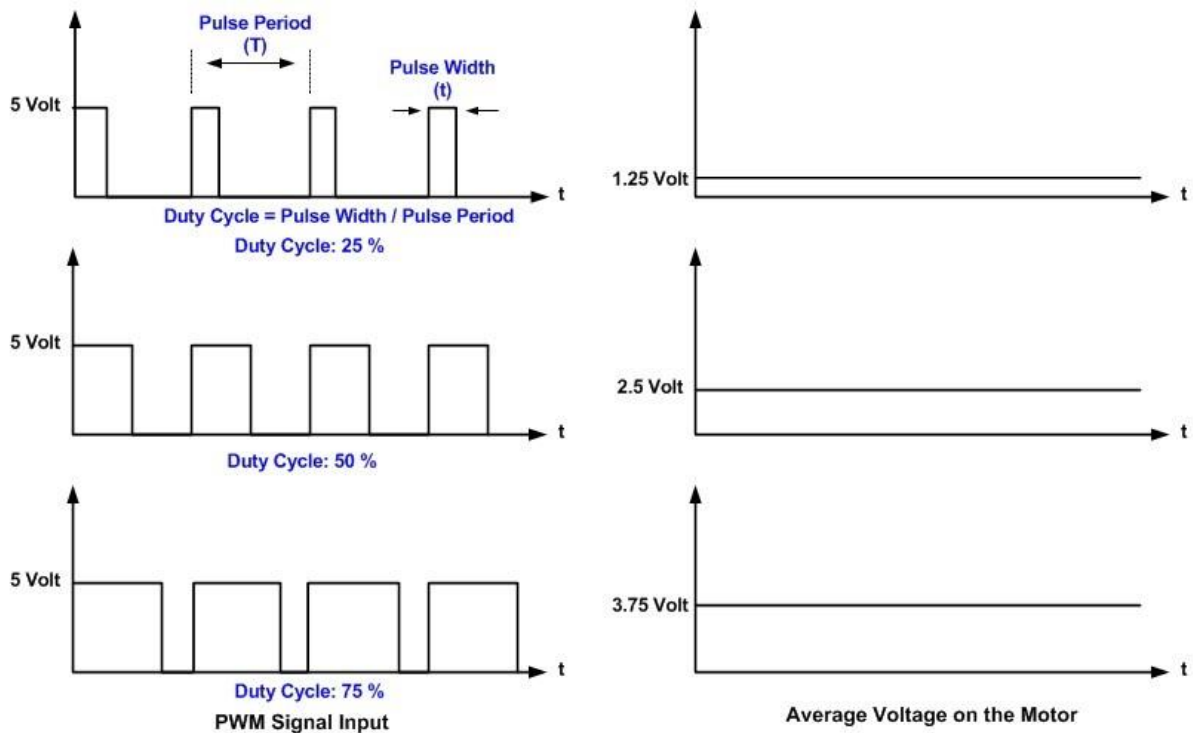


Fig 4.4: L298N Motor driver.

From the PWM timing diagram above we could see that by changing the pulse width we could change the average voltage receipt by the DC motor; the wider the pulse width; the higher the average voltage receipt by the DC motor. The shorter the pulse width, the lower the average voltage receipt by the DC motor. Therefore by varying the pulse width we could vary the DC motor speed. The ratio between the pulse width and the total length of the pulse (time on plus time off) is called duty cycle, so by saying 100% duty cycle means the DC motor is in it's full speed and 10% duty cycle the DC motor is in it's 10% of speed.

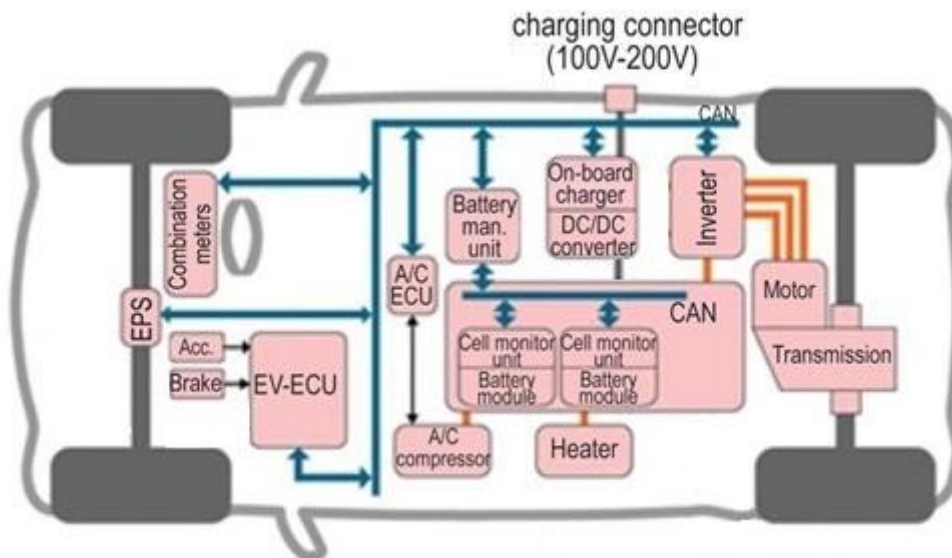


Fig 4.5: Basic Electric vehicle system configuration.

The basic electric vehicle configuration is shown in figure 4.5. The major electrical subsystems of an electric vehicle are controlled by each dedicated Electronic Control Unit (ECU). The main ECU is the EV-ECU, which controls the most important part of the vehicle, the drivetrain. The EV-ECU has accelerator, brake pedal inputs, and inputs from various other sensors in the vehicle.

Controller Area Network (CAN Bus) is a vehicle bus standard designed to allow microcontrollers and devices to communicate with each other in applications without a host computer. It is a message-based protocol, designed originally for multiplex electrical wiring within automobiles, but is also used in many other contexts.

Development of the CAN bus started in 1983 at Robert Bosch GmbH. The protocol was officially released in 1986 at the Society of Automotive Engineers (SAE) conference in Detroit, Michigan. The first CAN controller chips, produced by Intel and Philips, came on the market in 1987.

The modern automobile may have as many as 70 electronic control units (ECU) for various subsystems. Typically the biggest processor is the engine control unit. Others are used for transmission, airbags, antilock braking/ABS, cruise control, electric power steering, audio systems, power windows, doors, mirror adjustment, battery and recharging systems for hybrid/electric cars, etc. Some of these form independent subsystems, but communications

among others are essential. A subsystem may need to control actuators or receive feedback from sensors. The CAN standard was devised to fill this need.

Each battery module is connected to a cell monitoring unit , all of which are connected to the CAN bus.

The EV-ECU controls the motor by controlling the inverter which supplies power to the motor. The control signals to the inverter is given via the CAN bus.

CHAPTER 5

SIMULATION

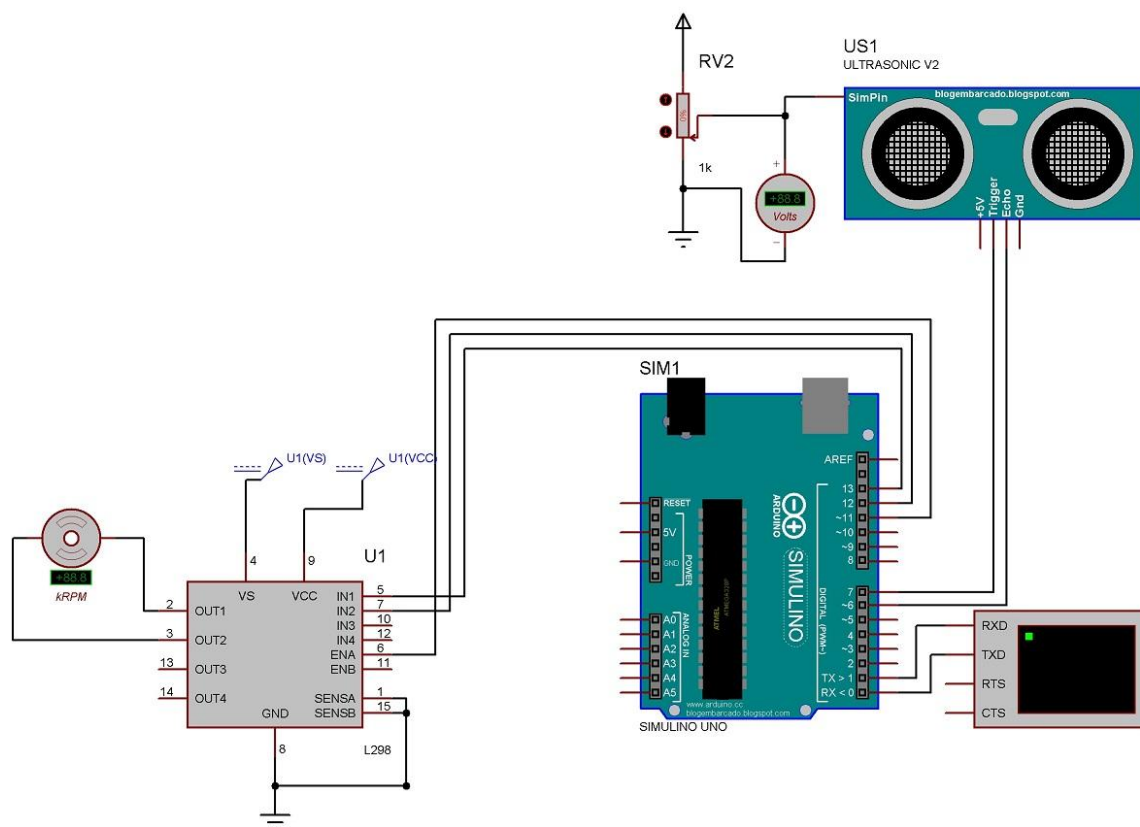


Fig 5.1: Simulation diagram.

The simulation has been completed using PROTEUS Software. The simulation diagram shown in Figure 5.1 consists of an ARDUINO UNO Microcontroller, an L298N Motor Driver and a HC-SR04 Ultrasonic Sensor . The motor of the vehicle cannot be directly interfaced with the microcontroller used here as the operating power of both the devices belong to different ranges. Hence we use an L298N Motor Driver to interface the motor to the microcontroller unit as it also supports PWM speed control of the motor. The input pins of the motor driver are manipulated to get different operating modes of the motor, depending upon which it rotates in a particular sense.

CHAPTER 6

HARDWARE PARTS

6.1 DISTANCE SENSOR – ULTRASONIC SENSOR

The Ultrasonic Sensor used is a HC-SR04 chip that is mainly used for sensing the presence of obstacles. In order to implement a working model of the proposed system, we are using a variable resistance in the circuit to introduce a voltage drop which represents the distance between the vehicle and the obstacle. This method has been used as it is not practical to introduce a physical obstacle in the circuit otherwise. As the variable resistance is varied, the presence of the obstacle is intensified with respect to the microcontroller unit and the speed of the motor connected decreases. When the distance is too small (that is the extreme end of the sliding contact of the variable resistance), the motor stops rotating because of the introduction of tight braking.

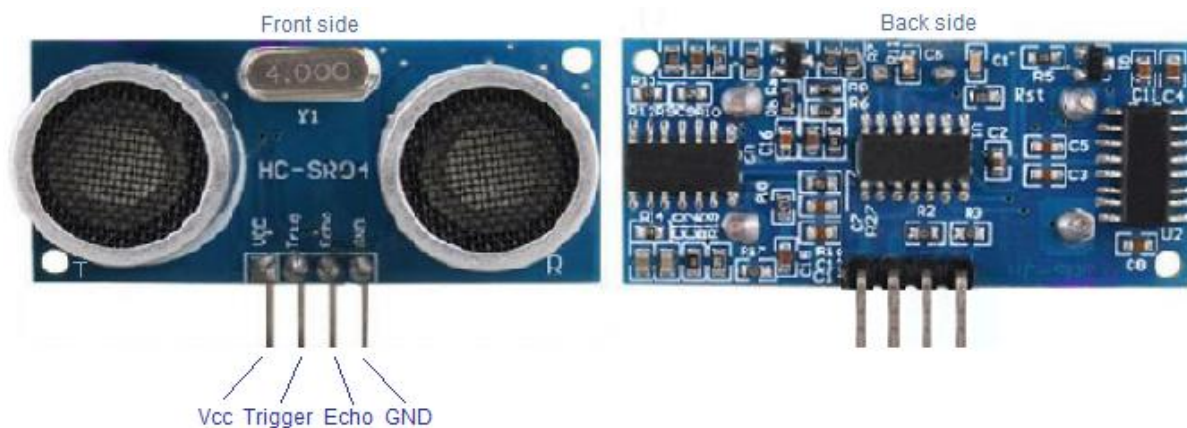


Fig 6.1: Ultrasonic sensor.

Ultrasonic ranging module HC - SR04 provides 2cm - 400cm non-contact measurement function, the ranging accuracy can reach to 3mm. The module includes ultrasonic transmitters, receiver and control circuit. The basic principle of work:

- (1) Using IO trigger for at least 10us high level signal,
- (2) The Module automatically sends eight 40 kHz and detect whether there is a pulse signal back.
- (3) IF the signal back, through high level , time of high output IO duration is the time from sending ultrasonic to returning.

Test distance = (high level time × velocity of sound (340M/S) / 2.

6.2 MICROCONTROLLER UNIT- ARDUINO UNO



Fig 6.2: ArduinoUNO board

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller.. It features the Atmega8U2 programmed as a USB-to-serial converter.

The Arduino Uno can be powered via the USB connection or with an external power supply. The power source is selected automatically.

External (non-USB) power can come either from an AC-to-DC adapter or battery. The adapter can be connected by plugging a 2.1mm center-positive plug into the board's power jack. Leads from a battery can be inserted in the Gnd and Vin pin headers of the POWER connector.

The board can operate on an external supply of 6 to 20 volts. If supplied with less than 7V, however, the 5V pin may supply less than five volts and the board may be unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

The Atmega328 has 32 KB of flash memory for storing code (of which 0,5 KB is used for the bootloader); It has also 2 KB of SRAM and 1 KB of EEPROM (which can be read and written with the EEPROM library).

The Arduino Uno can be programmed with the Arduino IDE (Intergrated development Environment) software

The ATmega328 on the Arduino Uno comes preburned with a bootloader that allows you to upload new code to it without the use of an external hardware programmer.

You can also bypass the bootloader and program the microcontroller through the ICSP (In-Circuit Serial Programming) header; see these instructions for details.

6.3 MOTOR DRIVER – L298N

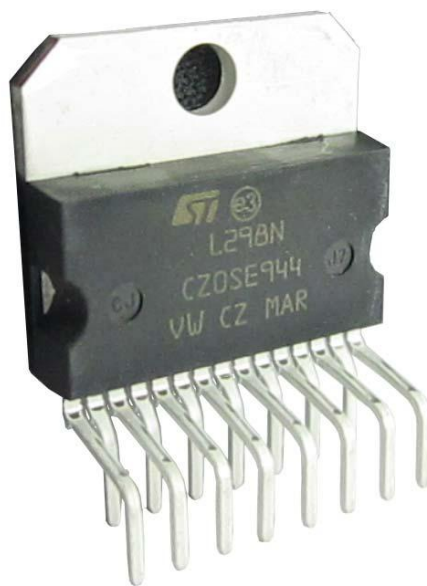


Fig 6.3: L298N Motor driver

INA	INB	EN	FUNCTION
1	0	1	FORWARD
0	1	1	REVERSE
1	1	1	FAST MOTOR STOP
0	0	1	FAST MOTOR STOP
X	X	0	FREE RUNNING STOP

Tab 6.1: L298N Operating Modes

The L298 is an integrated monolithic circuit in a 15- lead Multiwatt and PowerSO20 packages. It is a high voltage, high current dual full-bridge driver designed to accept standard TTL logic levels and drive inductive loads such as relays, solenoids, DC and stepping motors. Two enable inputs are provided to enable or disable the device independently of the input signals. The emitters of the lower transistors of each bridge are connected together and the corresponding external terminal can be used for the connection of an external sensing resistor. An additional supply input is provided so that the logic works at a lower voltage.

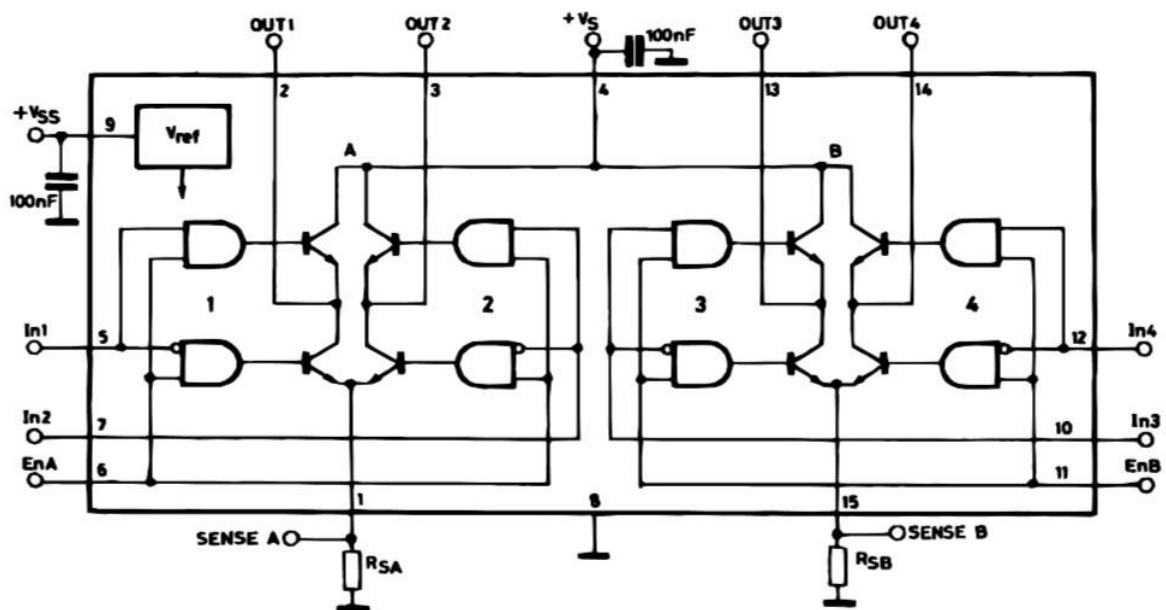


Fig 6.4: L298N Motor driver circuit.

The motor driver has 3 pins to control a single motor – IN1 IN2 and EN. These pins are connected to the output pins of the arduino board. The IN1 and IN2 pins are used to specify forward, reverse or braking. The EN pin is used for speed control and is connected to an output pin which supports PWM. If a logic high is given to EN pin, the motor will run at full speed. But if a PWM signal is given to the EN pin with a particular duty cycle, the motor will run at a speed corresponding to the duty cycle.

CHAPTER 7

PROGRAMMING

The software needed to run the control process of this system was developed using Arduino IDE software. The Arduino integrated development environment (IDE) is a cross-platform application written in Java, and derives from the IDE for the Processing programming language and the Wiring projects. It includes a code editor with features such as syntax highlighting, brace matching, and automatic indentation, and is also capable of compiling and uploading programs to the board with a single click. A program or code written for Arduino is called a "sketch". Arduino programs are written in C or C++.

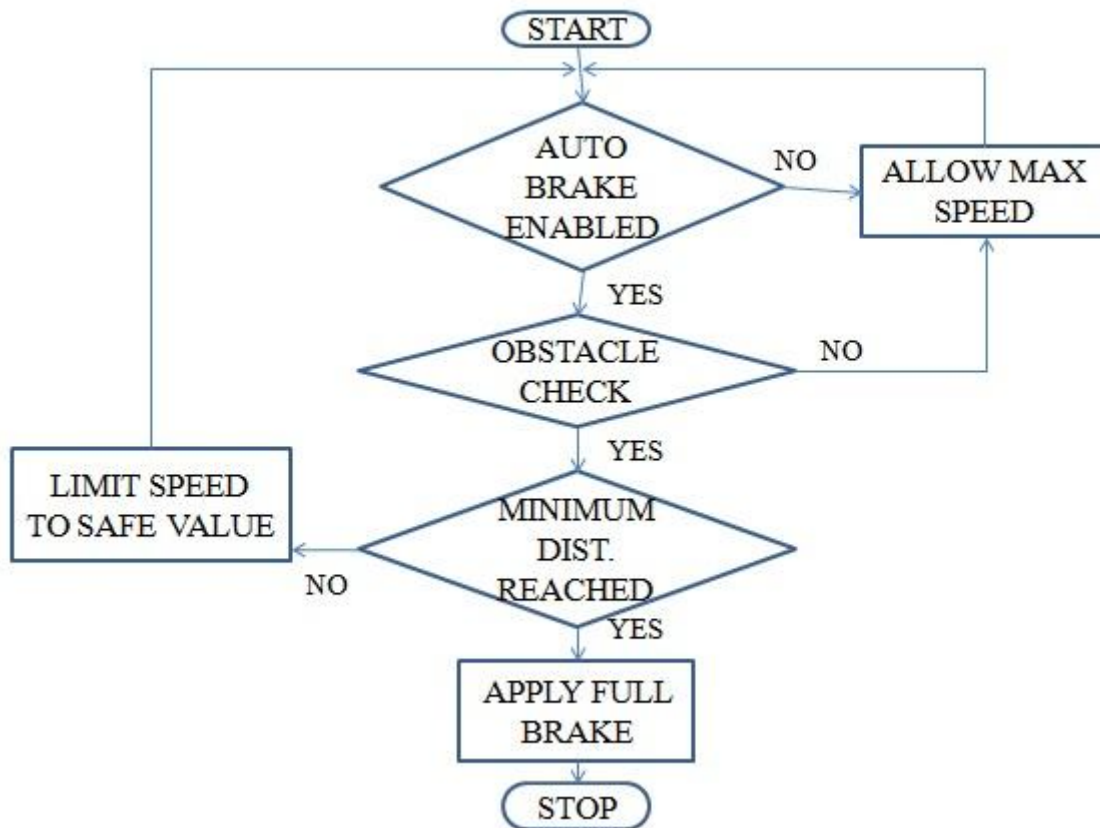


Fig 7.1:Flowchart of the program.

The behaviour of the system is completely based on the program uploaded to the control unit. The basic flowchart of the program used for the hardware prototype is shown in Figure 7.1.

Firstly the program checks if the system is enabled or not. If it is disabled maximum speed is allowed. Otherwise it checks for obstacles and if present, the vehicle speed will be limited to a safe value corresponding to the distance of the obstacle. If the obstacle is too close or if a threshold distance has been reached the vehicle applies full brakes and comes to a stop.

The program can be modified to any extent to suit the application depending on the size, type and purpose of the vehicle. The minimum distance for the vehicle to apply full braking, the amount of speed reduction, rate of deceleration, braking pressure, etc can be varied according to the needs.

CHAPTER 8

HARDWARE PROTOTYPE

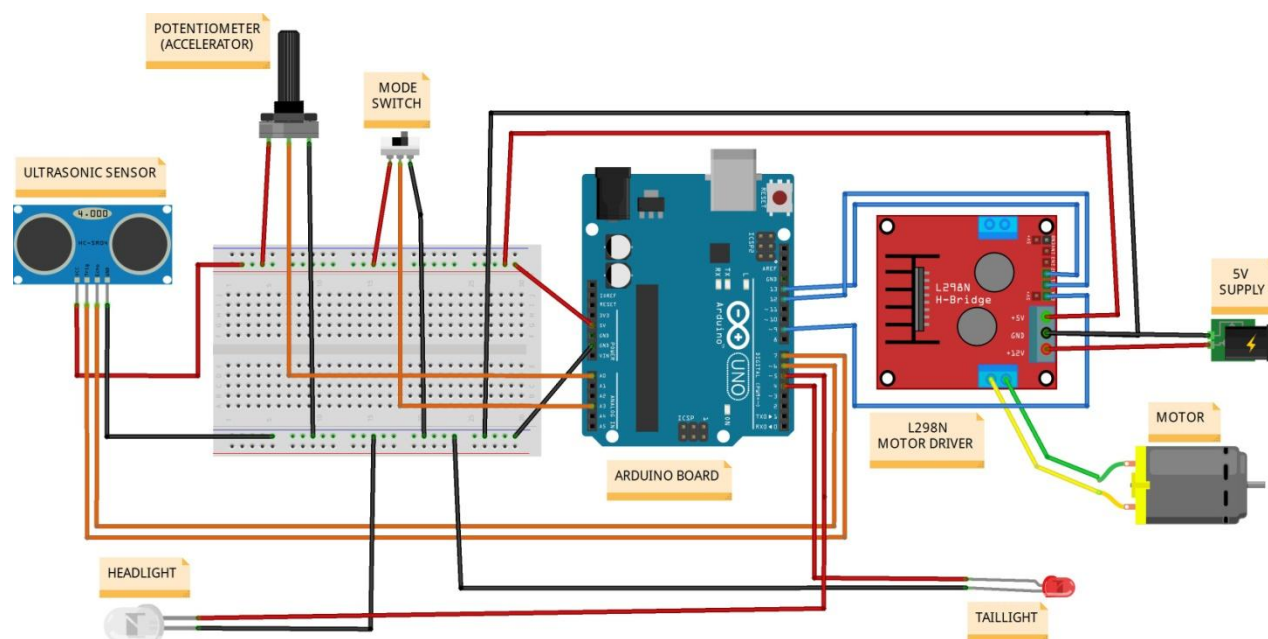


Fig 8.1: Circuit diagram of hardware prototype.

The microcontroller used here Arduino Uno. A potentiometer is provided to accelerate and decelerate vehicle. The analog output voltage coming from the potentiometer is given to the input of the Arduino board. The voltage from the potentiometer is read as an analog value and this value ranges from 0 – 1023 in the arduino board.

A mode switch is provided to enable or disable the system, the output of the switch is connected to the input of arduino board. The output of the switch is read as a digital value by the arduino board. The motor of the vehicle is interfaced to the Arduino Uno Microcontroller using an L298N Motor Driver. The motor driver has 3 pins to control a single motor – IN1 IN2 and EN. These pins are connected to the output pins of the arduino board. The IN1 and IN2 pins are used to specify forward, reverse or braking. The EN pin is used for speed control and is connected to an output pin which supports PWM. If a logic high is given to EN pin, the motor will run at full speed. But if a PWM signal is given to the EN pin with a particular duty cycle, the motor will run at a speed corresponding to the duty cycle. The duty cycle of the PWM produced by the arduino ranges from 0 – 255 corresponding to 0% and 100% duty cycle respectively. The analog value of the potentiometer output is then mapped to the duty cycle by

the arduino to run the motor at a speed corresponding to the current position of the accelerator pedal.

An Ultrasonic Sensor has been connected to the Microcontroller to detect the presence of obstacles, the sensor has 4 pins – Vcc,GND,Trig and Echo.The Trig pin of the sensor is connected to the output pin of the arduino and the Echo pin is connected to the input of the arduino.

The Arduino board has a 9V adapter as the power supply and it has an on-board 5V voltage regulator.The motor used here is a 5V motor so the power supply to the motor driver is from a 5V adapter.The logic power for the L298N motor driver is taken from the arduino board.

It has been equipped with Warning Lights to notify the driver the reducing distance between the vehicle.

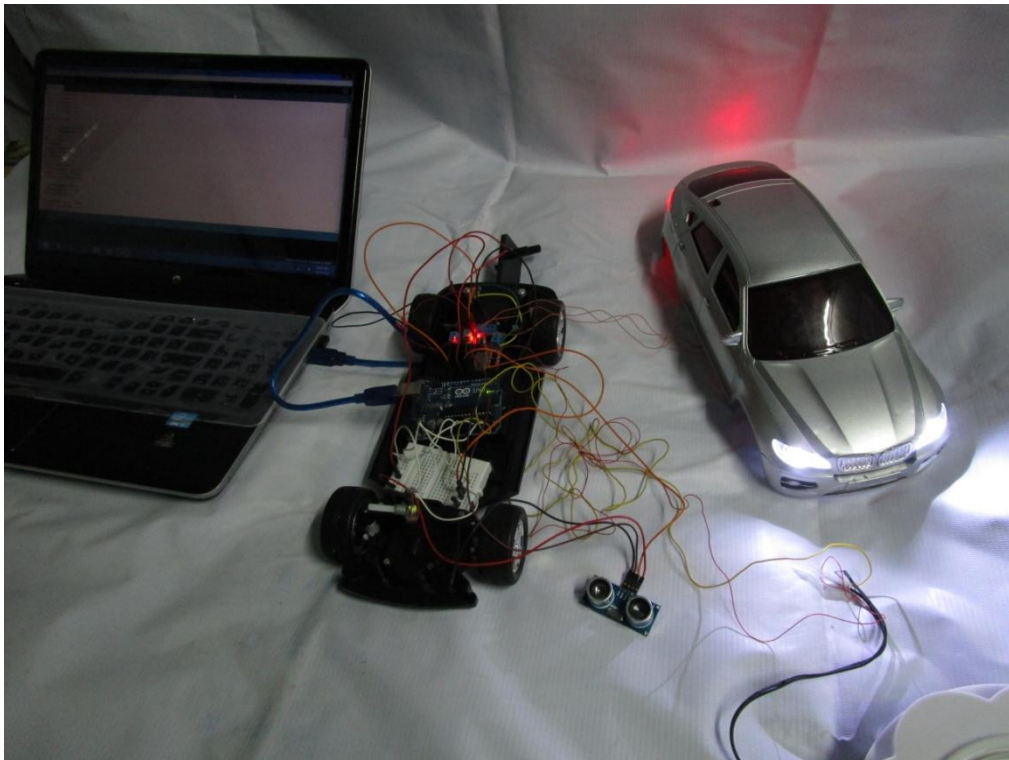


Fig 7.1:Hardware during testing.

CHAPTER 8

HARDWARE RESULTS

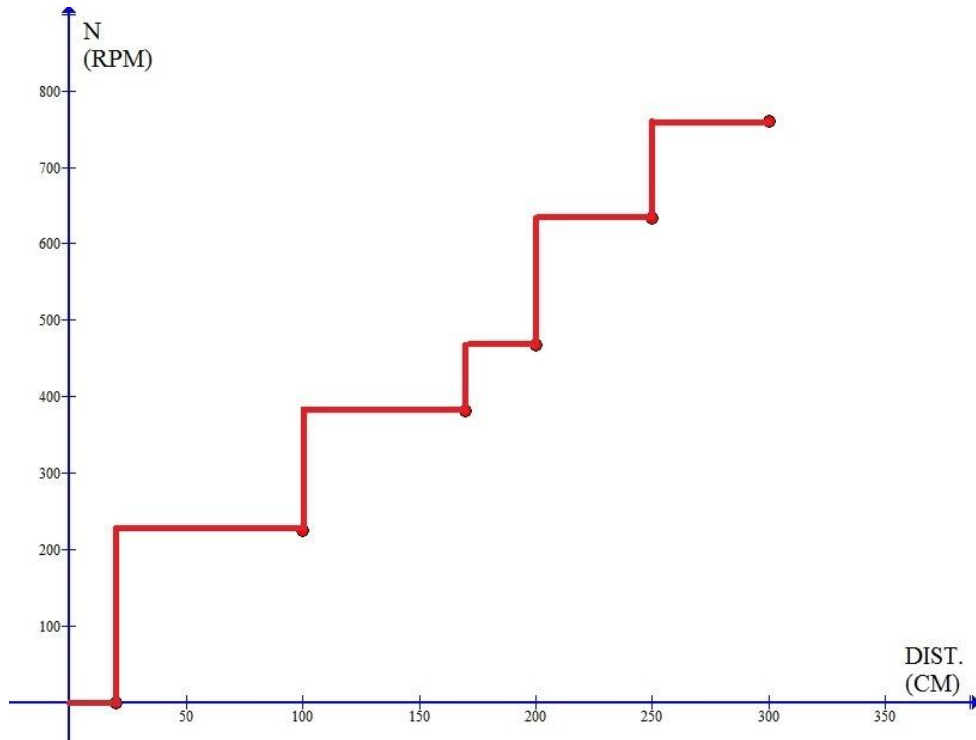


Fig 7.2: Hardware prototype results – Speed limit in rpm versus obstacle distance.

The prototype was tested and when the system was enabled the speed was reduced depending on the distance of the obstacle from the vehicle and imposed a speed limit. The vehicle was brought to a complete stop when the obstacle came too near to the vehicle. This is shown in Figure 7.2.

CHAPTER 5

ADVANTAGES

By the implementation of this system the stopping distance of the vehicle can be considerably reduced as the system has very fast response time. It can avoid rear end collisions of vehicles due to driver negligence. In the case of unavoidable collisions the system reduces the degree of damage. In control engineering aspect, the behaviour of the system can be changed by modifying the program uploaded to the electronic control unit.

CHAPTER 5

FUTURE SCOPE

Rain sensors and day night sensors can sense the environment conditions the visibility and road conditions. In wet road conditions the friction coefficient between the tyres and the road decreases so the system can automatically further reduce the speed to new safe values. Also the seat belt pretensioners can be connected to the system. Pretensioner is to tighten up any slack in the seatbelt in the event of a crash to make sure a seatbelt restrains an occupant as early as possible in a crash, thereby reduce the load on the occupant in a violent crash

CHAPTER 5

CONCLUSION

The advent of technology mainly focuses on reducing human effort in every field. This system may be considered as a step to reduce the mental and physical efforts taken by the driver to safely drive the vehicle. Even if our system may not prevent collisions in every critical scenario, it will definitely reduce the impact of collisions preventing loss of life. As the system can be made flexible, it can be suited for all kinds of vehicles and to the different traffic rules existing in different countries.

By the proper implementation of this system, the number of accidents occurring every year can be reduced by a considerable amount.

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APPENDIX

PROGRAM

```
const int modesw = A3;
const int trigPin = 7;
const int echoPin = 6;
const int headlgt = 4;
const int taillgt = 5;
int m;
long duration,cm;

const int analogInPin = A0;
const int analogOutPin = 9;
const int in1 = 12;
const int in2 = 13;
int safespd;
int sensorValue = 0;
int outputValue = 0;
void setup()
{
  pinMode(headlgt, OUTPUT);
  pinMode(taillgt, OUTPUT);
  digitalWrite(headlgt,HIGH);

  pinMode(echoPin, INPUT);
  pinMode(trigPin, OUTPUT);
  pinMode(modesw,INPUT);

  pinMode(analogInPin, INPUT);
  pinMode(analogOutPin, OUTPUT);
  pinMode(in1, OUTPUT);
  pinMode(in2, OUTPUT);
}
void loop()
```

```
{  
  m=digitalRead(modesw);  
  if(m==0)  
  {  
    digitalWrite(in1, HIGH);  
    digitalWrite(in2, LOW);  
    sensorValue = analogRead(analogInPin);  
    outputValue = map(sensorValue, 0, 1023, 0, 255);  
    analogWrite(analogOutPin, outputValue);  
  }  
  else if(m==1)  
  {  
    digitalWrite(in1, HIGH);  
    digitalWrite(in2, LOW);  
  
    digitalWrite(trigPin, LOW);  
    delayMicroseconds(2);  
    digitalWrite(trigPin, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(trigPin, LOW);  
  
    duration = pulseIn(echoPin, HIGH);  
    cm = microsecondsToCentimeters(duration);  
  
    if(cm<20)  
    {  
      rept:  
      digitalWrite(in1, LOW);  
      digitalWrite(in2, LOW);  
      analogWrite(analogOutPin, 255);  
      digitalWrite(taillgt,HIGH);  
      goto rept;  
    }  
    else if(cm<100)
```

```
{
    safespd=50;

}
else if(cm<170)
{
    safespd=100;
}
else if(cm<200)
{
    safespd=150;
}
else if(cm<250)
{
    safespd=180;
}
else if(cm<300)
{
    safespd=225;
}
else
{
    safespd=255;
}
sensorValue = analogRead(analogInPin);
outputValue = map(sensorValue, 0, 1023, 0, safespd);
analogWrite(analogOutPin, outputValue);
}
delay(50);
}
long microsecondsToCentimeters(long microseconds)
{
    return microseconds / 29 / 2;
}
```